COHOMOLOGICAL DESCENT FOR FPPF HYPERCOVERINGS ON QUASI-COHERENT MODULES: F -> RA_*A^*F IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the flatness identification La_{fppf}^* = a_{fppf}^* is justified by a_{fppf}^{-1}O_big = O_big,total (source lines 7660-7664, referring to remark-augmentation-ringed); this lemma upgrades card L46172 (equivalence on QCoh) to cohomological descent for the derived pushforward, over the diagram established on card L46173; no verification is omitted.

Editorial note. The label lemma-cohomological-descent-qcoh-for-fppf-hypercovering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The fppf-etale comparison diagram of topoi. Same chapter, \label{lemma-compare-simplicial-objects-fppf-etale} (card L46173). Proof lines 7631-7632 and 7650-7652: the commutative diagram with h, a_{fppf}, h_{-1}, a used throughout the computation.
- Componentwise cohomological descent for QCoh modules. More on Cohomology of Spaces, \label{spaces-more-cohomology-lemma-cohomological-descent-etale-fppf-modules}. Proof lines 7635-7638 and 7667-7673: F_n = a_n^*F satisfies F_n = Rh_{n,*}h_n^*F_n, and the closing equality Rh_{-1,*}(h_{-1})^*F = F.
- Direct image on the total site is componentwise (modules). Same chapter, \label{lemma-direct-image-morphism-simplicial-sites-modules}. Proof lines 7639-7640: a^*F = Rh_*h^*a^*F obtained by assembling the component identities.
- Simple descent for bounded modules along fppf hypercoverings. Same chapter, \label{lemma-hypercovering-X-simple-descent-bounded-modules}. Proof lines 7655-7660: Ra_{fppf,*}a_{fppf}^* = id with La_{fppf}^* = a_{fppf}^* since a_{fppf} is flat (a_{fppf}^{-1}O_big = O_big,total, remark-augmentation-ringed).

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Source reference. Original reference 0DAF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5910–5910. Prerequisite pointer invoked at proof line(s) 7664; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DH5: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7233–7233. Prerequisite pointer invoked at proof line(s) 7642; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DH1: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1656–1656. Prerequisite pointer invoked at proof line(s) 7649; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DAC: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5226–5226. Prerequisite pointer invoked at proof line(s) 7665; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D9W: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 4073–4073. Prerequisite pointer invoked at proof line(s) 7669; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DE7: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6985–6985. Prerequisite pointer invoked at proof line(s) 7638; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DGP: sources/stacks-chow/tags.txt, source lines 16879–16879. Prerequisite pointer invoked at proof line(s) 7647; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let U be a simplicial algebraic space over S. Let $a : U \to X$ be an augmentation. If $a : U \to X$ is an fppf hypercovering of X, then for $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module the map*

$$\mathcal{F} \to Ra_*(a^*\mathcal{F})$$

is an isomorphism. Here $a : Sh(U_{étale}) \to Sh(X_{étale})$ is as in Section 0DE7.

Proof. Consider the diagram of Lemma 0DH5. Let $\mathcal{F}_n = a_n^*\mathcal{F}$ be the nth component of $a^*\mathcal{F}$. This is a quasi-coherent $\mathcal{O}_{U_n}$-module. Then $\mathcal{F}_n = Rh_{n,*}h_n^*\mathcal{F}_n$ by More on Cohomology of Spaces, Lemma 0DGP. Hence $a^*\mathcal{F} = Rh_*h^*a^*\mathcal{F}$ by

Lemma 0DH1. We have

$$\begin{aligned} Ra_*(a^*\mathcal{F}) &= Ra_*Rh_*h^*a^*\mathcal{F} \\ &= Rh_{-1,*}Ra_{fppf,*}a_{fppf}^*(h_{-1})^*\mathcal{F} \\ &= Rh_{-1,*}(h_{-1})^*\mathcal{F} \\ &= \mathcal{F} \end{aligned}$$

The first equality by the discussion above, the second equality because of the commutativity of the diagram in Lemma 0DAF, the third equality by Lemma 0DAC as U is a hypercovering of X in $(Spaces/S)_{fppf}$ and $La_{fppf}^* = a_{fppf}^*$ as a_{fppf} is flat (namely $a_{fppf}^{-1}\mathcal{O}_{big} = \mathcal{O}_{big,total}$, see Remark 0D9W), and the last equality by the already used More on Cohomology of Spaces, Lemma 0DGP. □